	Risks and Opportunity Management	
	Document no: ASMF 8.5 Proc	
Compiled by:	Tshepo Maponya - Laboratory Supervisor	
Approved by:	Nonhlanhla Msibi - Operations Manager	

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Scope

This procedure establishes the operational framework for managing risks and opportunities. It ensures that technical validity, impartiality, and management system integrity are maintained across all laboratory activities, specifically fulfilling the requirements of **ISO/IEC 17025 (8.5)**, **ISO 17034 (8.5)**, and **ISO 17043 (8.5)**.

Document Control

Document No.	**PR-004-LAB**	**Effective Date**	**March 2026**
Associated Form	**ASRI 8.5-001** (Risk Register)	**Revision No.**	02

1. Operational Triggers for Risk Identification

The laboratory identifies risks and opportunities through five primary operational channels. When a risk is identified in any of these areas, it must be documented immediately in the **Excel Risk Register (ASRI 8.5-001)**.

1.1. **Operational Occurrences:** Technical failures such as equipment malfunction (e.g., uncalibrated balances), environmental drift in storage, or sample preparation errors (e.g., unlabelled samples).

1.2. **Client Interaction:** Formal complaints, appeals regarding test results, or feedback from PT scheme participants.

1.3. **Internal & External Audits:** Non-conformities or observations identified during technical or management system audits.

1.4. **Management Reviews (MRM):** Strategic risks identified during the annual review of the QMS, including resource adequacy and commercial pressures.

1.5. **Brainstorming & Scenario Analysis:** Proactive identification using SWOT analysis or flow charts to predict potential failures in new methods or projects.

2. Standardization Across Governing Regimes

To ensure compliance, the risk assessment must address the specific requirements of the three governing standards either as standalones or interchangeably:

- **ISO/IEC 17025:** Focus on technical validity, metrological traceability (e.g., calibration schedules), and laboratory impartiality.
- **ISO 17034:** Focus on the integrity of Certified Reference Materials (CRMs), specifically risks to homogeneity and long-term stability.
- **ISO 17043:** Focus on the validity of Proficiency Testing (PT) schemes, including risks of participant collusion and late report delivery.

3. Documentation Process (ASRI 8.5-001)

The following steps dictate how a risk is moved from identification to the formal register:

3.1. **Entry:** The identifying staff member records the **Identified Risk**, the **Potential Cause**, and the **Consequence** (e.g., "Results for time-sensitive parameters affected" or "Lack of metrological traceability").

3.2. **Category Assignment:** Categorize the risk (e.g., Impartiality, Methodological, Financial, or Health & Safety).

3.3. **Ownership:** Assign a **Risk Owner** (e.g., Laboratory Manager or Operations Manager) responsible for the treatment plan.

4. Assessment and Rating Methodology

Each risk is quantified in the Excel Register using the laboratory's standard probability and effectiveness criteria.

4.1. Likelihood Criteria (Probability)

| Category | Criteria |

| **E** | approx 99% probability; occurring now or within days/weeks. |

| **D** | > 50% probability; likely to occur within weeks/months. |

| **C** | > 20\% probability; may occur within months/years. |

| **B** | > 5\% probability; not anticipated but possible in years/decades. |

| **A** | < 5% probability; exceptionally unlikely; "100-year event." |

4.2. Risk Control Effectiveness (RCE)

The existing controls are evaluated to determine if further action is required:

- **Fully Effective (F):** Controls are preventative, address root causes, and are always reliable.
- **Partially Effective (P):** Controls are in place but require improvement in operational reliability.
- **Ineffective (I):** Controls do not treat root causes or rely too heavily on reactive measures.
- **None (N):** No credible control exists.

5. Impartiality and Conflict of Interest

5.1. Risks to impartiality are assessed on an ongoing basis.

5.2. Personnel must record any "Dual Roles" or "Commercial Pressure" in the register either during onboarding inductions or continuously.

5.3. Controls must be implemented to ensure that remuneration or financial dependence on specific clients does not influence the technical outcome of PT rounds or CRM certification.

6. Risk Treatment and Tasks

6.1. For any risk where the RCE is rated as **Partially Effective** or lower, a **Task to Improve Control** must be documented.

6.2. Tasks must include a **Task Owner** and a **Due Date**.

6.3. Typical treatments include:

- **Technical:** Decommissioning uncalibrated equipment or updating sampling plans.
- **Administrative:** Implementing interim controls (e.g., hold-and-release points) for unlabelled samples.
- **Resource-based:** Addressing insufficient equipment (e.g., analytical balances) to meet turnaround times.

7. Monitoring and Continual Improvement

7.1. The Risk Register (ASRI 8.5-001) is reviewed during every Management Review Meeting to ensure all tasks are completed.

7.2. Lessons learned from successful opportunities (e.g., improved communication flow) are integrated into the standard operating procedures to enhance the effectiveness of the QMS.

7.3. Records of risk assessment and treatment are maintained for a minimum of 5 years to provide evidence of conformity during SANAS assessments.

Revision History

Revision No	Date	Content Changed	Reason